

Second Preliminary Report

ECLift

XtraChallenge 2025

March 1st 2025

Project Overview

Since the beginning of the 2024 academic year, the ECLift team has made significant progress in the aircraft design and fabrication for the XtraChallenge 2025. The initial phases focused on team organization, conceptual design, and material characterization. As of now, all major design aspects are finalized, and the team is fully engaged in the manufacturing stage. This report summarizes our developments in structural fabrication, aerodynamic finalization, avionics integration, and upcoming testing phases.

Aircraft Design and Configuration

Figure 1 provides an isometric view of the skinned aircraft. The aircraft features a fully composite skin for both the fuselage and internal wing sections, optimizing load transfer efficiency. The mid-wing, puller, V-tail configuration was chosen to maximize climb rate, extend glide time, and support a maximum payload capacity of approximately 2L. These design choices informed the aerodynamic philosophy, resulting in a mid-wing placement, a V-tail, and a high-aspect-ratio wing. Figure 2 presents the planform of the main wing. Figure 3 depicts the finalized tail planforms. Stability analysis, shown in Figure 4, confirms a stable, negatively sloped static stability curve when carrying a payload. This finalized configuration informed the preliminary mass distribution illustrated in Figure 5.

Structural Configuration

The structural design leverages composite materials to minimize weight while maximizing strength throughout the airframe. The internal framework, seen in Figure 6, is optimized to accommodate a carbon-skinned fuselage, reducing the need for excessive internal formers and stringers. The tail assembly is detachable for streamlined payload loading, comprising a connector plate, carbon tube, and V-tail mounting slots. The inner wing sections feature composite skins with unidirectional carbon spar flanges laid up during fabrication, followed by a post-bonded wooden shear web. Wing panels connect via short carbon tubes, balancing structural integrity with ease of assembly. The outer wing panels and V-tail are constructed using composite spars, wooden internals, and an Oracover skin to minimize cost.

Propulsion and Avionics

The propulsion layout has been designed to ensure organized internal cabling paths. The aircraft's nose houses primary electronics, with large connectors extending to control surfaces. Servo installations are distributed among the wing panels, with access facilitated by well-sized hatches. The redesigned thrust test stand, depicted in Figure 7, features an extensometer and LED screen. Thrust testing is scheduled to commence shortly.

Manufacturing Process

For composite sections, plugs and parting boards were essential for mold production. These components were 3D printed in sections over several weeks, then assembled, primed, and sanded to achieve a high-quality surface finish. A tooling gel coat and thick laminate were applied to create durable negative molds, this process can be seen in Figure 8. Top-bottom molds were preferred for their symmetry, particularly for airfoil and fuselage sections. Currently, molds have been completed, and composite layups will begin soon.

For composite skin sections, fiberglass-carbon fiber layups are prioritized, though fiberglass-balsa-fiberglass compositions will be tested for comparison and training purposes. Non-composite skin sections will feature custom laminate spars made from unidirectional carbon fibers, combined with wooden ribs, spacers, and shear webs. To facilitate Oracover application, structural spacers will be strategically placed. The team is committed to best practices in composite manufacturing, emphasizing proper fiber handling, vacuum curing, and precision sealing. With significant progress underway, the first prototype is expected by mid-March.

Challenges and Solutions

The primary challenges encountered involved mold manufacturing. Efforts to minimize costs led to thinner laminate layers, resulting in weaker than expected molds. This issue has been mitigated through post-repair modifications. Another major hurdle has been the lack of flight experience. To date, the club has yet to conduct a successful flight test, prompting increased focus on flight readiness in the coming weeks.

Next Steps and Testing

As we progress into the final phase before flight testing, the upcoming objectives include : completing final wing-fuselage assembly, conducting structural load testing on wings and payload holders, performing preliminary taxi and ground tests, and initiating full-scale flight validation.

Conclusion

ECLift has successfully transitioned from design to manufacturing. With all major components being fabricated, the focus now shifts to integration and testing. The team remains on track for a competitive performance at the upcoming XtraChallenge 2025.

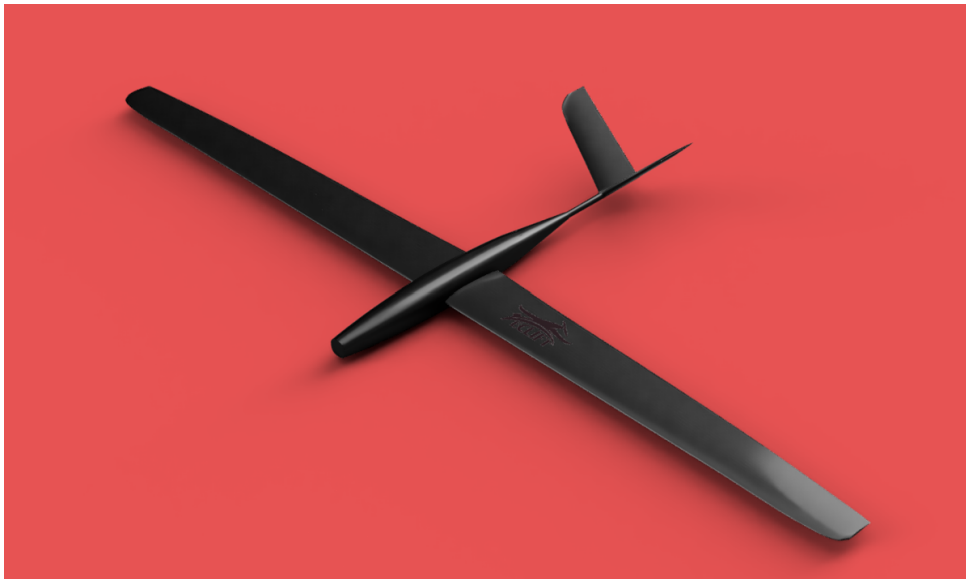


FIGURE 1 – Isometric view of the skinned aircraft

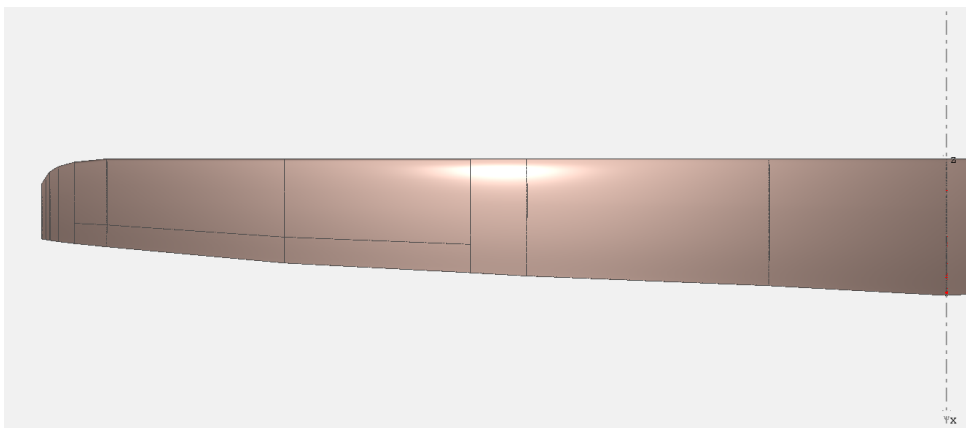


FIGURE 2 – Planform of the main wing

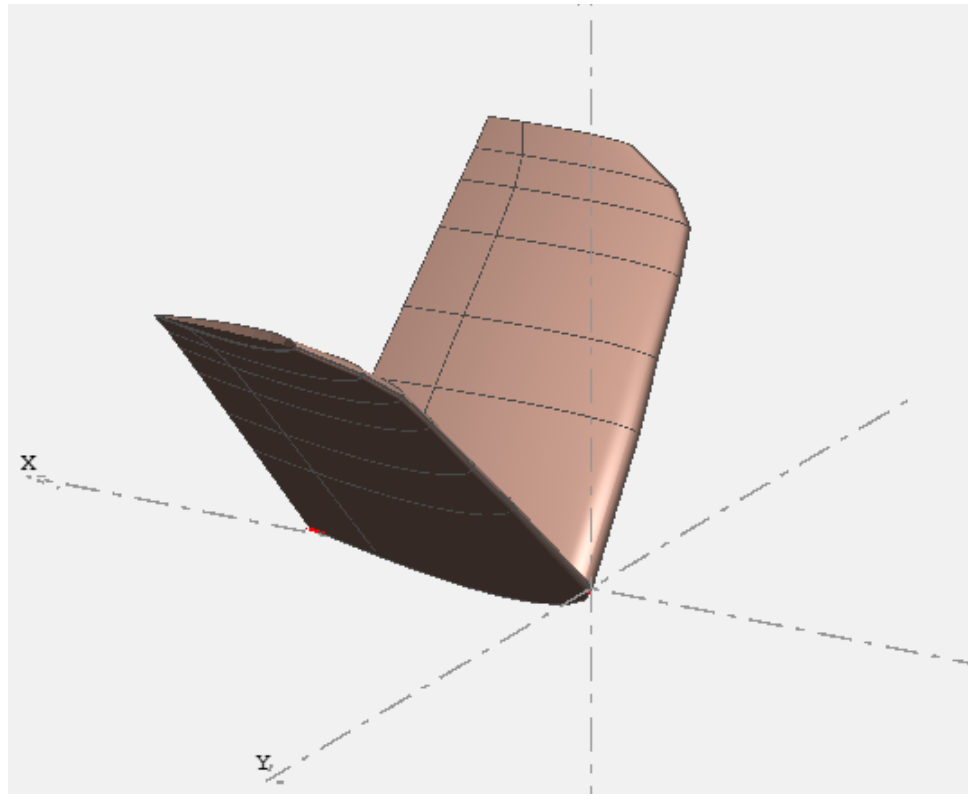


FIGURE 3 – Isometric view of the V-Tail

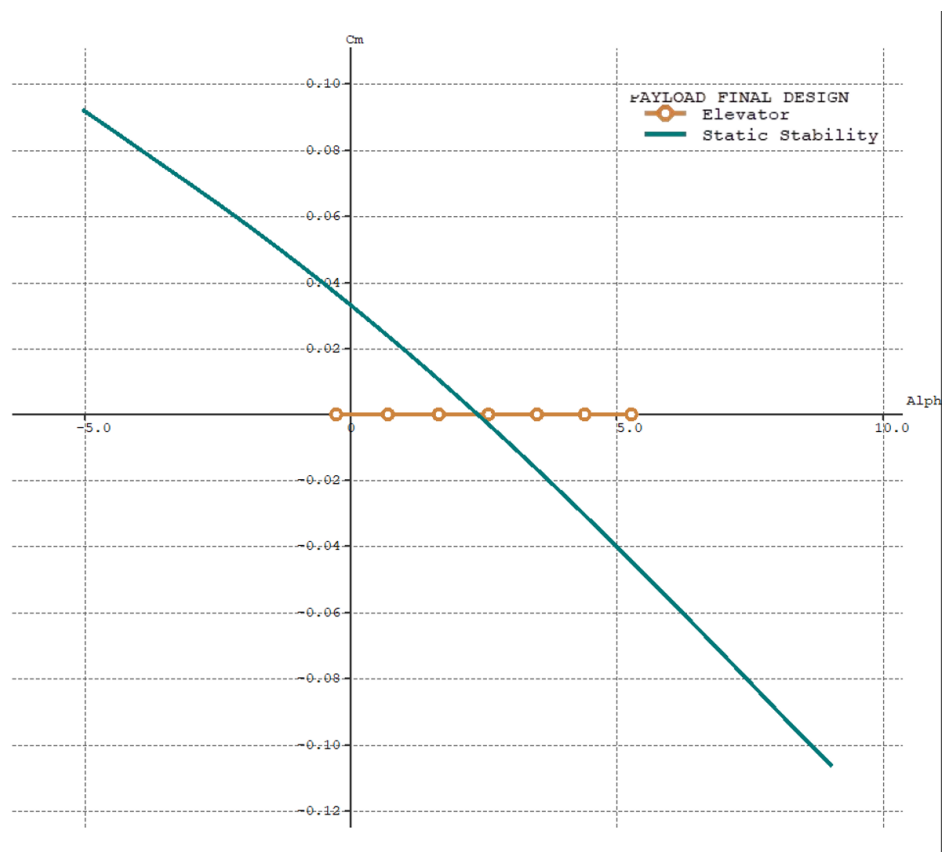


FIGURE 4 – Static stability analysis with elevons

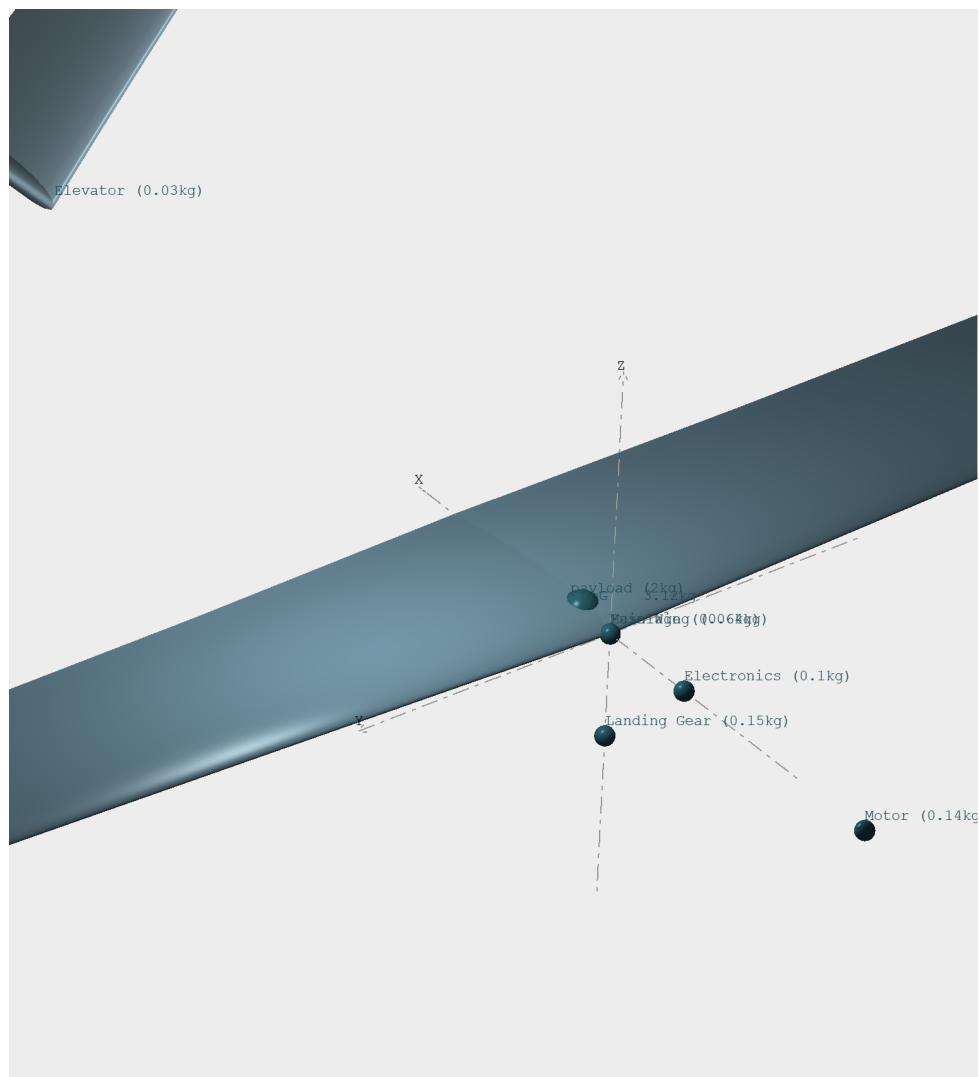


FIGURE 5 – Preliminary mass distribution

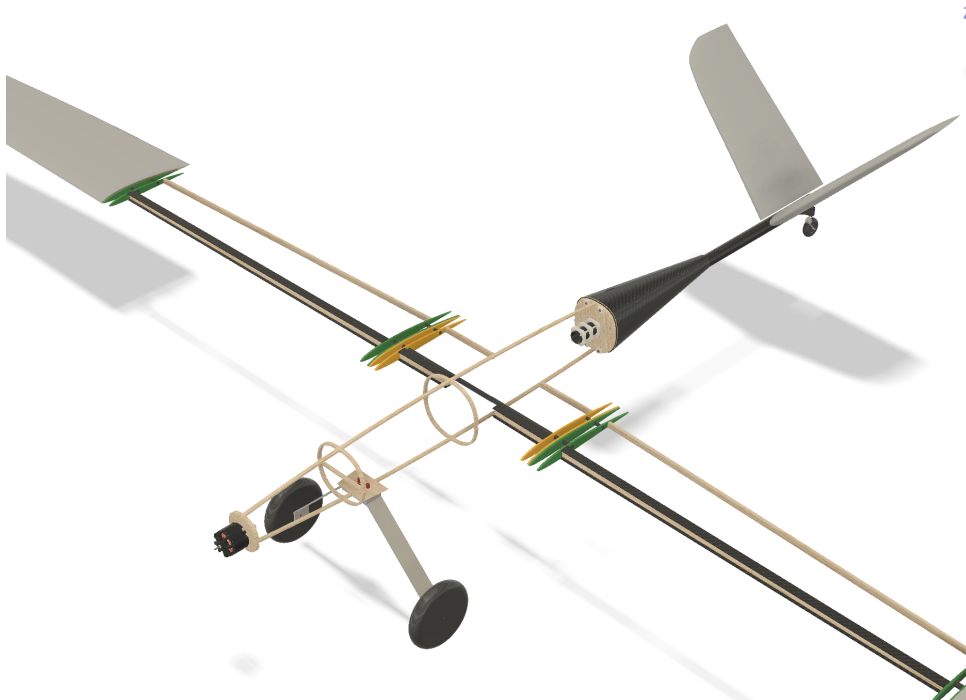


FIGURE 6 – Internal framework of the air frame

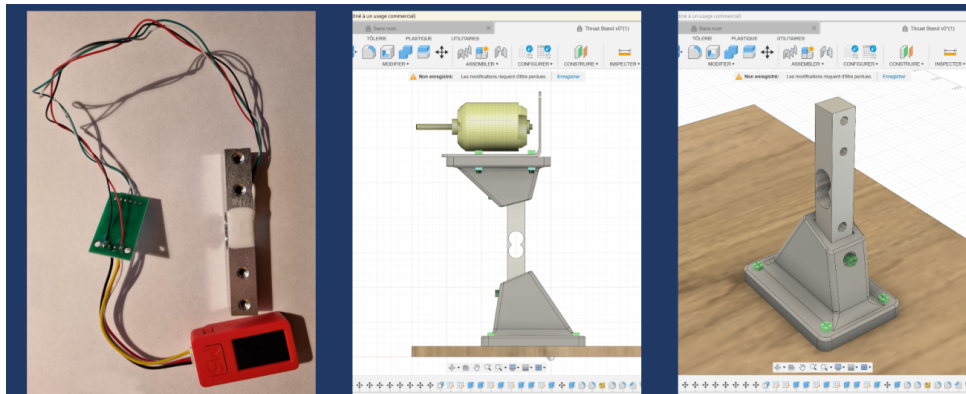


FIGURE 7 – Redesigned thrust test stand



FIGURE 8 – Mold manufacturing process